

Web Dev & Python

A Beginner's Guide for Dummies

1. HTML - The Skeleton

HTML (HyperText Markup Language) is the backbone of every website. Think of it like the skeleton of a body — it gives structure to web pages.

Key Concepts:

- Tags: HTML uses tags like `<h1>`, `<p>`, `<img>` to define content
- Elements: A tag + its content = an element. E.g., `<p>Hello</p>`
- Attributes: Extra info on tags. E.g., ``
- Structure: Every HTML page has `<html>`, `<head>`, and `<body>`
- Nesting: Tags go inside other tags, like folders inside folders

```
<!DOCTYPE html>
<html>
  <head><title>My Page</title></head>
  <body>
    <h1>Welcome!</h1>
    <p>This is my first web page.</p>
  </body>
</html>
```

2. The `<div>` Tag - The Box

A `<div>` is like an invisible box you use to group content together. Think of it like a shipping container — you can put anything inside it and move it around as one unit.

Why use divs?

- Group related elements together (a card, a section, a sidebar)
- Apply CSS styles to the whole group at once
- Create layout structure (columns, rows, grids)
- Target sections with JavaScript

```
<div class="card">
  <h2>Product Name</h2>
  <p>$29.99</p>
  <button>Buy Now</button>
</div>
```

Think: div = "division" of the page. Each div is a section you can style and control independently.

3. CSS - The Clothing

CSS (Cascading Style Sheets) controls how HTML looks — colors, sizes, fonts, spacing, layouts. If HTML is the skeleton, CSS is the skin and clothes.

```
.card {
  background-color: white;
  border-radius: 8px;
  padding: 20px;
  box-shadow: 0 2px 4px rgba(0,0,0,0.1);
}
```

4. JavaScript - The Brain

JavaScript (JS) makes web pages interactive. It's the brain that responds to clicks, updates content, fetches data, and controls behavior. Every modern website uses it.

Key Concepts:

- Variables: Store data. `let name = "Bryn"; const age = 20;`
- Functions: Reusable blocks of code. `function greet() { ... }`
- Events: React to user actions. `button.addEventListener("click", ...)`
- DOM: JavaScript can change HTML on the fly (add/remove/modify elements)
- Arrays & Objects: Store lists and structured data
- if/else: Make decisions. `if (score > 90) { grade = "A"; }`
- Loops: Repeat actions. `for (let i = 0; i < 5; i++) { ... }`

```
// Change text when button is clicked
const btn = document.getElementById("myBtn");
btn.addEventListener("click", () => {
  document.getElementById("output").textContent = "Hello!";
});
```

Analogy: HTML = the house structure, CSS = the paint and decor, JavaScript = the electricity and plumbing that makes things work.

5. Python - The Swiss Army Knife

Python is a general-purpose programming language known for being easy to read and learn. It's used for web backends, data science, AI, automation, and more.

Key Concepts:

- Variables: `name = "Bryn"` (no `let/const` needed)
- Indentation matters! Python uses spaces instead of `{ }` braces
- Lists: like arrays. `fruits = ["apple", "banana", "cherry"]`
- Dictionaries: key-value pairs. `student = {"name": "Bryn", "grade": "A"}`
- Functions: `def greet(name): return f"Hello {name}"`
- Loops: `for item in fruits: print(item)`
- Libraries: `import math, import pandas` — thousands of add-ons

```
# A simple Python program
def calculate_average(grades):
    total = sum(grades)
    return total / len(grades)
my_grades = [85, 92, 78, 95, 88]
avg = calculate_average(my_grades)
print(f"Your average is {avg}")
# Output: Your average is 87.6
```

Quick Reference Cheat Sheet

HTML !' Structure (what's on the page)

CSS !' Style (how it looks)

JS !' Behavior (what it does)

div !' A box to group HTML elements

Python !' Backend logic, scripts, data, AI